

EXERCISE 2

STREAM CIPHERS

1. We will now analyze a pseudorandom number sequence generated by a Linear Feedback Shift Register (LFSR) of degree 3 characterized by $(p_2 = 1, p_1 = 0, p_0 = 1)$.

- (1) What is the sequence generated from the initialization vector: $(s_2 = 1, s_1 = 0, s_0 = 0)$?
- (2) What is the sequence generated from the initialization vector: $(s_2 = 0, s_1 = 1, s_0 = 1)$?
- (3) How are the two sequences related?

2. Assume we have a stream cipher whose period is quite short. We happen to know that the period is 150–200 bits in length. We assume that we do not know anything else about the internals of the stream cipher. In particular, we should not assume that it is a simple LFSR. For simplicity, assume that English text in ASCII format is being encrypted.

Describe in detail how such a cipher can be attacked. Specify exactly what Oscar has to know in terms of plaintext/ciphertext, and how he can decrypt all ciphertext.

3. Compute the first two output bytes of the LFSR of degree 8 and the feedback polynomial $(0, 1, 7)$ where the initialization vector has the value FF in hexadecimal notation.

4. In this problem, we will study LFSRs in more detail. LFSRs come in three flavors:

- LFSRs which generate a maximum-length sequence. These LFSRs are based on primitive polynomials.
- LFSRs which do not generate a maximum-length sequence, but whose sequence length is independent of the initial value of the register. These LFSRs are based on irreducible polynomials that are not primitive. (Note that all primitive polynomials are also irreducible.)
- LFSRs which do not generate a maximum-length sequence and whose sequence length depends on the initial values of the register. These LFSRs are based on reducible polynomials.

We will study examples in the following. Determine all sequences generated by the following three polynomials:

- (1) $x^4 + x + 1$
- (2) $x^4 + x^2 + 1$
- (3) $x^4 + x^3 + x^2 + x + 1$

Draw the corresponding LFSR for each of the three polynomials. Which of the polynomials is primitive, which is only irreducible, and which one is reducible? Note that the lengths of all sequences generated by each of the LFSRs should always add up to $2^m - 1$.

5. Given is a stream cipher based on a single LFSR as the keystream generator. The LFSR has a degree of 256.

- (1) How many plaintext/ciphertext bit pairs are needed to launch a successful attack?
- (2) Describe all steps of the attack in detail and develop the formulae that need to be solved.
- (3) What is the key in this system? Why doesn't it make sense to use the initial contents of the LFSR as the key or as part of the key?

6. We conduct a known-plaintext attack on an LFSR-based stream cipher. We know that the plaintext sent was:

1001 0010 0110 1101 1001 0010 0110

By tapping the channel, we observe the following stream:

1011 1100 0011 0001 0010 1011 0001

- (1) What is the degree m of the key stream generator?
- (2) What is the initialization vector?
- (3) Determine the feedback coefficients of the LFSR.
- (4) Draw a circuit diagram and verify the output sequence of the LFSR.

7. We want to perform an attack on another LFSR-based stream cipher. In order to process letters, each of the 26 uppercase letters and the numbers 0, 1, 2, 3, 4, 5 are represented by a 5-bit vector according to the following mapping:

$$\begin{aligned}
 A &\leftrightarrow 0 = 00000_2 \\
 &\vdots \\
 Z &\leftrightarrow 25 = 11001_2 \\
 0 &\leftrightarrow 26 = 11010_2 \\
 &\vdots \\
 5 &\leftrightarrow 31 = 11111_2
 \end{aligned}$$

We happen to know the following facts about the system:

- The degree of the LFSR is $m = 6$.
- Every message starts with the header WPI.

We observe the following message on the channel (the fourth symbol is a zero): j5a0edj2b

- (1) What is the initialization vector?
- (2) What are the feedback coefficients of the LFSR?
- (3) Write a program in your favorite programming language which generates the whole sequence, and find the whole plaintext.
- (4) Where does the thing after WPI live?
- (5) What type of attack did we perform?

8. In this problem, we will look at pseudorandom number generators based on a linear congruential generator (LCG). As we have seen in Section 2.2.1, an LCG is given by the equations:

$$\begin{aligned} z_0 &\equiv \text{seed} \\ z_{i+1} &\equiv a \cdot z_i + b \pmod{m}, \quad i = 0, 1, \dots \end{aligned}$$

We assume that the modulus m is public and that the key is formed by the parameters seed , a , and b .

We consider now the problem that arises if we use an LCG as a keystream generator. We assume the stream cipher is used for encrypting images given in GIF format. The keystream z_i encrypts a plaintext x_i as follows:

$$y_i \equiv x_i + z_i \pmod{m}$$

Since GIF files consist of 8-bit values, we need at least 256 possible values for x_i . Thus, the prime modulus $m = 257$ is a good choice.

Now, assume that the first six bytes in the header of a GIF image file consist of the letters GIF89a, where each letter is encoded as an 8-bit ASCII character. An attacker who obtains an encrypted GIF file finds the following ciphertexts at the beginning of the file: $y_1 = 32$, $y_2 = 166$, and $y_3 = 87$, which correspond to the plaintext bytes $x_1 = \text{G}$, $x_2 = \text{I}$, and $x_3 = \text{F}$.

- (1) Describe how an attacker can compute the parameters a , b , and the seed with these three plaintext bytes. Compute the parameters a , b , and the seed.
- (2) What is this attack called? What are the prerequisites for a successful attack?

9. The linear congruential generator described in Section 2.2.1 can be extended such that each new element z_{i+1} of the keystream is computed from the two previous elements z_i and z_{i-1} . In this case, two seed values z_0 and z_1 as well as three parameters a , b , and c along with the modulus m are needed. The equation of the generator is given as:

$$z_{i+1} \equiv a \cdot z_i + b \cdot z_{i-1} + c \pmod{m}$$

The keystream z_i is used to encrypt letters given in 8-bit ASCII code on a character-by-character basis. That means for each keystream value z_i , one plaintext character x_i is encrypted as:

$$y_i \equiv x_i + z_i \pmod{m}$$

Oscar, the attacker, eavesdrops on the communication and happens to know that the ciphertext contains the name ALICE, starting at position i . Oscar observes the following ciphertext symbols on the channel:

$$y_i = 69, \quad y_{i+1} = 47, \quad y_{i+2} = 3, \quad y_{i+3} = 88, \quad y_{i+4} = 217$$

He also knows that the modulus $m = 257$ is being used.

Show how Oscar can compute the parameters a , b , and c .

10. Assume the initialization vector (IV) and the key of Trivium each consist of 80 all-zero bits. Compute the first 70 bits $s_1, \dots, s_{70}$ during the warm-up phase of Trivium. Note that these are only internal bits, which are not used for encryption since the warm-up phase lasts for 1152 clock cycles.